Ratcheting Elliptic-Curve Index Calculus Toward Pollard Rho:

A Ledger of Verified Improvements, the Distance That Remains,

and the Claims That Did Not Survive (July–October 2026)

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Abstract

Pollard rho remains the fastest known algorithm for the discrete logarithm on a prime-order elliptic curve. Index calculus, which can in principle exploit more structure, has never beaten it on curves of cryptographic shape. Between July and October 2026 we treated closing that gap as a *ratchet*. Each verified, measured improvement over the current best index-calculus pipeline became the new baseline, and every row reports its remaining distance to a matched rho. This paper is the ledger of that work, written from committed evidence and current to 23 October 2026.

We organise the improvements by the stage they act on.

- *Decomposition oracle.* Root-finding and meet-in-the-middle oracles are 383–1,535× cheaper per decomposition than matrix F4, and up to $10^5\times$ cheaper than the earlier point-decomposition oracle. Inherited F4 does 199–222× less matrix work than matrix F4. An $\mathbb{F}_p$ F4 reaches msolve parity. A gate-first order and a root-rank harvest gate each cut the Gröbner work further, by 1.3–1.7× and 1.41×.
- *Factor base and relation yield.* Free relations (0.960×), re-randomised targets (0.531×), a greedy coverage base, and a confirmed $n = 83$ base selection (1.195 s of 115 s cold).
- *Set-up and linear algebra.* A stacked $n = 83$ producer (3.24× cold), a cache-blocked pair table (0.24× set-up), and CADO-style sparse linear algebra (10.4×).
- *Two families whose fitted cost exponent lies below rho's.* Genus-three index calculus after the cubic GHS transfer, whose whole solve moved from $q^{1.91}$ to a local $q^{1.36}$ over five rounds. Gaudry's $k = 4$ index calculus with FGHR's D_4 symmetry, 370–481× cheaper per decomposition, whose gap to rho closes as $n^{-1/4}$ from 5,080× to 6.1×.

The distance remaining is stated plainly. On Koblitz curves of 36.6–47.2 bits, against a matched rho, the cold gap is 4.67–18.07×. On prime curves it is 10.5× at 20 bits and grows to 102× at 40. The nearest families are genus three (1.15× a measured rho at $q = 2^{20}$) and $k = 4$ in characteristic two (4.3× a measured rho at $n \approx 2^{56}$, one target). Neither is a deployed curve shape.

We also report the claims that did not survive. All online wins over rho are retired: Bernstein–Lange preprocessing rho with equal advice is 53–317× faster. Pricing native work and matching the rho reference each raised a recorded gap, and a replay regression was traced to a cross-ISA counter divergence. As of this writing no index-calculus leaderboard row has passed the repository's validation gate. Nothing here is a speed-up on a deployed curve or a key recovery.

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1 Introduction

Let $\langle G \rangle$ be a subgroup of prime order r of an elliptic curve $E/\mathbb{F}_q$. The best generic algorithm for its discrete-logarithm problem is Pollard rho [14, 16]. When A efficiently computable automorphisms act on the subgroup, rho costs about $\sqrt{\pi r/(2A)}$ group additions [7, 18]. Index calculus was brought to elliptic curves through Semaev's summation polynomials [15] and the extension-field analyses of Gaudry and Diem [4, 8]. It is the method that should exploit the most structure. On prime-order curves of cryptographic shape it has never beaten rho.

This paper does not ask whether index calculus beats rho; at every size we can measure, it does not. It records how far a sustained engineering programme moved index calculus *toward* rho, one verified step at a time, and how far remains.

The success criterion. On 15 October 2026 the repository's owner fixed the rule that governs this paper (`AGENTS.md`, "Success criterion: iterative improvement of the index-calculus baseline"):

- the baseline for each curve and size is the best current verified index-calculus pipeline, frozen with its protocol and receipts;
- an *improvement* lowers cost against that baseline on the same instances, with every answer verified, under a protocol frozen before timing;
- each landed improvement becomes the new baseline (the ratchet);
- every row reports its gap to the matched rho as the distance remaining, not as a pass/fail gate;
- relabelling and accounting changes are recorded but are not improvements.

Beating rho remains the long-term goal, and a claim to have done so still needs a cold, end-to-end, independently replayed win at growing sizes. No such claim appears here.

Relation to the programme record. A companion record ([1], `papers/ecdlp-index-calculus-program/`) surveys the literature, the measurement infrastructure and the programme's chronology through 20 October. This paper takes a narrower view in three ways:

1. it organises the work by the pipeline stage each change acts on (§§4–7), and states each change against its own incumbent;

2. it collects the distance to rho by curve family, as it stands after the 19–20 October corrections, including the matched-rho re-run of the Koblitz rule (§8);
3. it adds the work of 20–23 October: the confirmed $n = 83$ base selection, the cache-blocked pair table, the planner validation, the ECCp-131 relation probes, the cross-ISA root cause of the replay regression, and the state of the validation gate (§§9–10).

Where the two papers overlap, this one cites the same committed artefacts. It ran no new measurement.

Contributions.

- A cost decomposition of elliptic-curve index calculus into the levers the programme actually pulled, and the share of total cost each one controls (§2).
- The ratchet: about thirty improvements, each against its own incumbent, with boundary, class, interval and source (Figure 1, Table 2).
- The distance to matched rho, by family and size, with the reference each figure uses (§8).
- An audit trail of claims that were corrected, and why each correction moved the gap *up* (§9).
- The state of the validation gate and what each row needs to pass it (§10).
- The next improvements, ranked by the stage that now dominates cost (§11).

What this paper does not claim.

- No ECDLP speed-up on any deployed curve, and no key recovery.
- No asymptotic sub-rho algorithm for prime fields, and no change to the Gaudry–Diem picture.
- No statement about ECC2K-130, P-256 or secp256k1 security.

Every instance is a public synthetic or standard-parameter known-answer instance. Stage ratios are never presented as end-to-end speed-ups. Wall times from hosts without a qualifying isolation level are labelled *descriptive*. A cost with unpriced work is labelled a *lower bound*.

Reading the numbers. We use two conventions.

- In the ledger (§§4–7) a ratio is *candidate/incumbent*, so less than 1 is an improvement. Where a source reports a speed-up s, we write it as $s\times$ and plot $1/s$.
- In distance statements (§8) a ratio is *IC/rho*, so greater than 1 means rho is cheaper.

Intervals are 95%: paired cluster-bootstrap or Student-t, as the source recorded them. Artefact paths are relative to the root of the `crypto` repository unless prefixed by a sibling repository's name.

2 Where the cost of index calculus lives

An improvement is only useful if it moves a term that matters. This section writes the cost of one index-calculus solve as a sum of terms, and maps each lever of the programme to the term it acts on.

2.1 The pipeline

Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$ of size N. A *relation* is a decomposition $aG + bQ = P_1 + \cdots + P_m$ with every $P_i \in \pm\mathcal{F}$. Once about N independent relations are known, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. A heuristic count is enough to place the levers. A uniformly random point decomposes into m base points with probability about $N^m/(m!\,r)$. Collecting N relations therefore needs about $m!\,r/N^{m-1}$ trials. The cold cost of one target is then

$$T \;\approx\; \underbrace{T_{\mathrm{setup}}(N)}_{\text{base, tables}} + \underbrace{\frac{m!\,r}{N^{m-1}}\,\frac{1}{\rho_{\mathrm{yield}}}\,C_{\mathrm{dec}}(N,m)}_{\text{relation collection}} + \underbrace{T_{\mathrm{LA}}(N,m)}_{\text{linear algebra}} + \underbrace{T_{\mathrm{target}}}_{\text{descent, check}} \quad . \tag{1}$$

Here C_{dec} is the cost of one decomposition attempt (the *oracle*). The factor $\rho_{\mathrm{yield}} \geq 1$ collects everything that makes a trial more productive than the heuristic: free relations, large primes, symmetry, re-randomised targets. T_{LA} is roughly mN^2 for sparse elimination. With rho's cost $\sqrt{\pi r/(2A)}$ on the other side, the classical balance at $m = 2$ with a constant-cost oracle gives $N \approx r^{1/3}$ and $T \approx r^{2/3}$. That is why two-summand prime-field index calculus loses ground to rho as $r^{1/6}$ (§8).

2.2 The levers

Table 1 lists, for each term of (1), the levers the programme pulled and the sections that report them.

Table 1: The terms of (1) and the levers that act on them.

Term	Levers pulled	Section
C_{dec}	root finding of S_3; pair and meet-in-the-middle tables; inherited F4, F5, F6-IC; gate-first order; harvest gates; $\mathbb{F}_p$ F4; D_4 invariants	§4, §7
ρ_{yield}, N	compact Frobenius orbits; free relations; stride-renewed targets; greedy coverage; $n = 83$ base selection; double large primes	§5, §7
T_{setup}	stacked producer; cache-blocked pair table; u64 root batching	§6
T_{LA}	CADO-style filter and Wiedemann; structured Gaussian elimination; block Wiedemann; first-dependency stop	§6, §7
the exponent	genus-three transfer (cubic GHS); $k = 4$ extension fields	§7

2.3 Which term dominates now

Early in the programme, C_{dec} dominated. A Gröbner solve per trial cost milliseconds to seconds. After root finding (§4), the oracle is a handful of field operations and table probes, and the cost has moved elsewhere.

- On the Koblitz descent of the leaderboard's Table A, reusable set-up is 98.5–99.7% of cold S, and descent is 0.3–1.4% (`docs/ic/LEADERBOARD.md`, phase table).
- On the $n = 83$ cold pipeline with base K1107, a mean 113.755 s process splits into 47.166 s of set-up and 66.486 s of rank stage. The target stages take 0.099 s (`research/n83_cold_base_confirm_20261021/RESULT.md`).

So the relation-collection and set-up terms dominate. Their cost is set by N and by the yield per probe, not by the oracle. This is the main structural fact behind the ledger. Oracle speed-ups of three to five orders of magnitude moved the end-to-end gap by one order at most.

3 How an improvement is measured

The measurement contract is set out in full in the programme record [1] (§3 there). This section gives only what is needed to read the ledger.

Unit. The common unit is S = (total group-addition equivalents, GAE)$/\sqrt{r}$, with floor $S_{\mathrm{floor}} = \sqrt{\pi/(2A)}$. An addition or doubling is one GAE. Native work is converted at pinned calibration ratios (`docs/ic/calibration.json`) where a conversion exists. Such work includes table lookups, row operations, Artin–Schreier solves and solver word XORs. Where no conversion exists, the work is recorded as unpriced and the cost is a lower bound. Some families use their own counted unit: $\mathbb{F}_q$ or $\mathbb{F}_p$ multiplications for genus three and $k = 4$, or a batched affine addition for the Koblitz rule. A ratio is comparable only within one unit.

Boundaries. Every number in this paper is stated on one of four boundaries:

- *stage*: one oracle call, one solver run, one set-up phase;
- *complete*: a verified discrete logarithm with the stated phases;
- *cold*: one previously unseen target, every table built inside the cost;
- *online*: one target after reusable preparation, which is excluded.

The success criterion's ratchet is defined on the complete or cold boundary. A stage improvement is reported as a stage improvement. It becomes a pipeline improvement only once it moves end-to-end cost.

Records. Measurements are made with the native harness *ecbench* (`src/cryptanalysis/ecbench/`) or a study's own frozen driver. Each execution appends a sealed record, and every answer is checked as $[k]G = Q$ outside the solver. `ecbench verify` re-executes measured records and requires identical operation counts; the SHA-256 of its receipt is the replay certificate. Wall time is admitted only at the isolation level its specification requires (default L2: quiet host, zero steal). Below that level it is descriptive.

The pricing band. Almost no cost is complete, the rho baselines included. Since 15 October, every cross-method comparison prices each arm's unpriced native work alike at $w \in \{0, 0.1, 1\}$ GAE per operation. It ranks two arms only if their order holds at all three weights (`AGENTS.md` §5a).

Change classes. Every change is classed as one of:

- ADVANCE: an exponent or crossover moves under complete accounting;
- ENGINEERING: a constant moves;
- RELABELLING: the headline count fell while total cost rose;
- ACCOUNTING: the numbers changed but the algorithm did not.

Against the index-calculus incumbent, the ledger also uses the verdicts of `ecbench`'s challenge protocol. A change *improves the exponent* (Ie) or *the constant* (Ic), is engineering on one stage (E), *trades* (T), or *regresses* (R). An Ie verdict against the incumbent is not an ADVANCE against rho unless the exponent crosses rho's.

The validation gate. Since 19 October no row may enter the index-calculus leaderboard or `benchmarks/README.md` without three pieces of evidence, named by the evidence directory's `VALIDATION.json` (`docs/VALIDATION_GATE.md`):

1. an independent replay certificate with verified scalars;
2. an independent validator report that recomputes the headline numbers;
3. a matched rho reference: the strongest available, in the same unit and the same binary, priced across the band.

Rows recorded before the gate render as "unvalidated (pre-gate)". §10 gives the state of every IC row.

4 The ratchet, I: the decomposition oracle

Figure 1 plots every improvement in the ledger as candidate/incumbent on a log scale. Table 2 (in §7) gives each one's class, boundary and source. This section covers the oracle, C_{dec} in (1).

4.1 Root finding replaces the Gröbner solve

For two summands, the decomposition question has a closed-form answer. For $E : y^2 = x^3 + ax + b$ and affine R, F with $x_R \neq x_F$,

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \big(X - x(R+F)\big)\big(X - x(R-F)\big).$$

So $R = \pm F_i \pm F_j$ if and only if $x(R \mp F_i) \in x(\mathcal{F})$. The identity is machine-checked in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`). The resulting oracle is one subtraction and one hash probe per base point, with every inversion batched. For $m = 3, 4$, a table of pair sums turns a decomposition into N or N^2 probes. The mechanism is the second algorithm of McGuire and Mueller [12]; what is new here is engineering and measurement.

Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $N = 55$) and 1,535× ($\mathbb{F}_{2^{13}}$, $N = 4{,}005$) cheaper than matrix F4. Every verdict agreed with F4. The `rf-mitm` oracle refutes a binary target 3,007–115,160× faster than the earlier PS-0 oracle across $\ell = 8..14$ (`research/pdp_root_finder_20261012/`). The meet-in-the-middle oracle on a non-Koblitz curve over $\mathbb{F}_{2^{37}}$ reached $\ell = 12$. It is 928–4,244× faster than pairs-and-solve on $\ell = 9..12$, with no disagreement on 256 targets (`research/binary_mitm_oracle_l11_l12_20261016/`).

The end-to-end effect is two orders of magnitude smaller than the stage effect:

- against msolve, on 16–24-bit prime curves, 50–105×;
- on the one binary instance large enough to amortise the table ($n = 23$), 122×; F4 is faster at $n \leq 13$;
- inside ecbench at Koblitz $n = 13$, a root-finding arm folded over negation costs 0.0545 [0.047, 0.064] of an inherited-F4 arm whose word XORs are priced (`research/ecbench_rootfind_faster_20261014/`).

A late correction affects the last comparison. On 20 October a cross-ISA replay showed that F4's counted word XORs depend on whether the host has BMI2 (§9). Once F4 is priced, the root-finding pair table is about 17–22× cheaper than F4 at $n = 13$ (`research/validation_sweep_20261019/D-known-issues/REPORT.md`). The direction is unchanged.

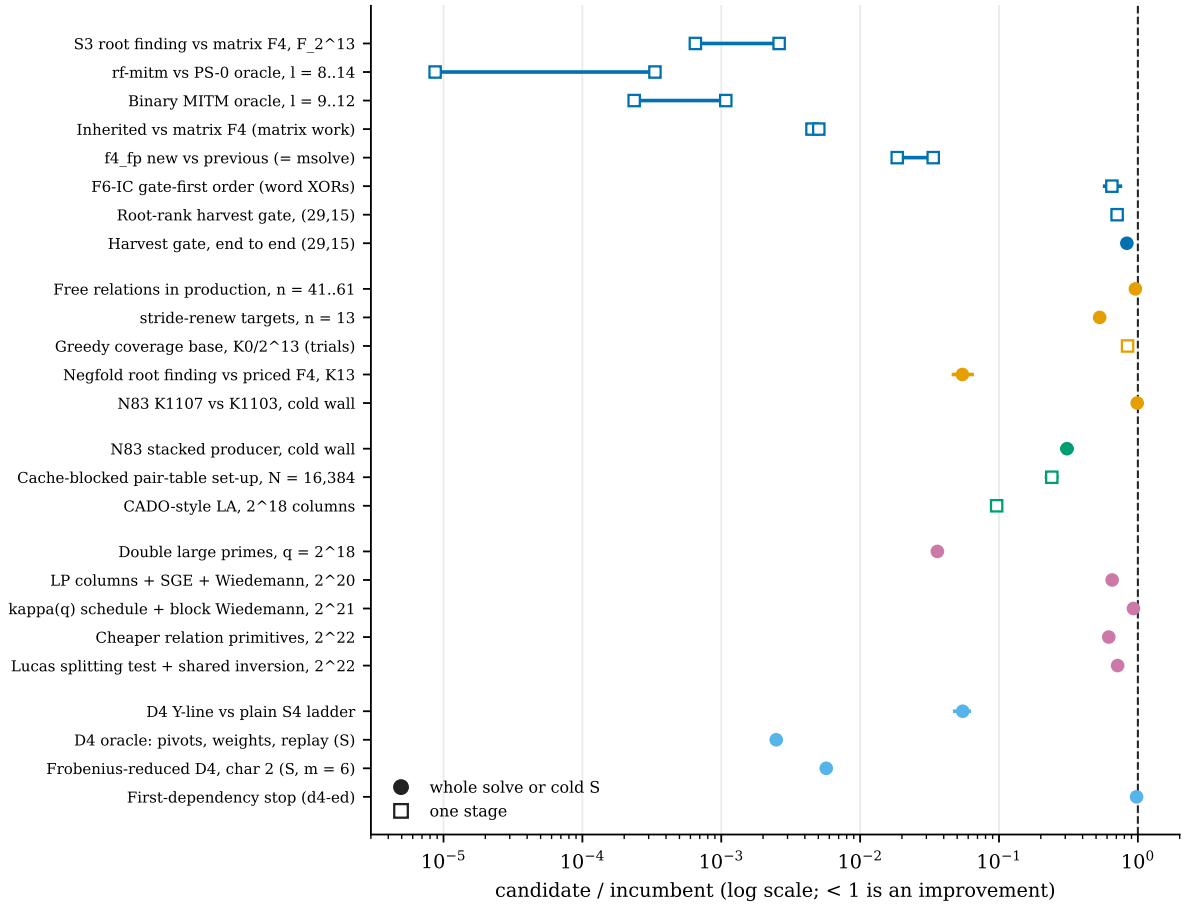


Figure 1: The ratchet. Each row is one change, candidate/incumbent on the study's own boundary; less than 1 is an improvement. Filled circles are whole solves or cold S; open squares are single stages. Bars show 95% intervals. Rows with two markers show the range a source reports across instances, not an interval. Rows are grouped by pipeline term and are not comparable across groups: incumbents, curves and units differ. Values and sources: `papers/ecdlp-ic-improvement-ledger/figures/data.json`.

4.2 The Gröbner family, where it is still needed

For $m \geq 3$ over binary fields the oracle is still a Gröbner solve. Four changes moved it.

- *Inherited F4.* Each child node of the DPLL-style tree inherits its parent's echelon form. This does 198.6–222.1× less matrix work than matrix F4 on identical trees. At $m = 19$ matrix F4 timed out at 1,800 s, while inherited F4 finished in about 6 s (`research/f4_f5_f6_cross_family_20261011/`).
- *$\mathbb{F}_p$ F4 at msolve parity.* Gebauer–Möller criteria, packed monomials and interned symbolic preprocessing made `f4_fp` 29.8–54.1× faster than its predecessor. msolve over ours is 0.96–1.22 single-threaded, and every reduced basis agrees with Singular or msolve (`research/f4_fp_msolve_parity_20261011/`). Blocked pair reduction then gave 0.78–0.89× wall on the quadratic $n = 10, 11$ systems (`research/f4_fp_blocked_pairs_20261016/`; L0, descriptive).
- *Gate-first order in F6-IC.* Checking the factor-base domain gate before reduction cuts word XORs by 1.336–1.733× and CPU by 1.56–2.12×. All 80 of 80 targets were decided identically (`research/f6_gate_first_x86_20261016/`). It is now the default. An arm64 rerun on 22 October found that turning it off costs 1.62–2.15×, so every default stands (`research/arm64_f4_defaults_20261022/README.md`; shared CI VMs, one run each).
- *Harvesting what the oracle discarded.* Linear rows that the Gröbner oracle used to throw away are kept: 0.858 $[0.803, 0.916]$ at $(n, \ell) = (23, 12)$, where the gate fires. A root-rank gate then made the harvest fire on more cells: 1.41× $[1.32, 1.52]$ CPU at $(29, 15)$, 1.89× at $(37, 19)$, and 1.20× and 1.14× end to end at $(29, 15)$ and $(31, 16)$, with 0 of 248 root-set failures (`research/ic_improve3_20261015/02-harvest-gate`).

Changes that did not count.

- *F6-IC as a frontier move.* F6-IC uses 0.77–0.82 of inherited F4's word XORs. Once its batched group additions are charged, though, its S is 64.6–1,353× inherited F4's. That is ACCOUNTING.
- *Algebraic oracles against the MITM floor.* No algebraic oracle (inherited F4, F6-IC, SAT, split-S_4 hybrids, a chained-S_3 solver) grows more slowly in n than the meet-in-the-middle floor. The cheapest is $3.0 \cdot 10^2\times$ the floor at $n = 13$ and $3.8 \cdot 10^4\times$ at $n = 31$ for $m = 3$ (`research/ic_m3_algebraic_vs_mitm_20261014/`).
- *Root-shape replay in inherited F4.* It regresses by 1.015–1.328×, so the switch stays off (`research/pdp_shape_replay_20261015/`).
- *F4 trace replay in the Jacobian-cover pipeline.* It now costs +31% and +35% (`research/ic_cover_leads_20261018/README.md`).

Reading. The oracle is the term the programme improved most: by two to five orders of magnitude where root finding applies, and by one to two orders in counted matrix work where Gröbner bases are still needed. None of this changes the trial count $m!\, r/N^{m-1}$. That is why the remaining sections deal with N, the yield, set-up and linear algebra.

5 The ratchet, II: factor base and relation yield

This section covers N and ρ_{yield} in (1). On Koblitz curves these terms are the main target, because the curve's Frobenius gives the factor base structure the index calculus can use.

5.1 Compact Frobenius orbits

On $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$ the Frobenius π acts on the subgroup as a scalar λ. A π-invariant base splits into orbits with $\log \pi^k(P) = \lambda^k \log P$, so the unknowns collapse by about n [6]. The programme's *compact signed-Frobenius orbit* base stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). It is the best measured Koblitz construction:

- in the ecbench factor-base suite at $n = 17$ it is 9.985× matched signed-Frobenius rho, against 18.7× for full orbits and 47× for subspace bases (`research/ecbench_frontier_20261013/`);
- in the priced frontier campaign it is 3.8–15× rho at $w = 0$ and 4.9–17.8× at $w = 0.1$ on eight Koblitz curves (`research/ic_frontier_campaign_20261015/`);
- fully priced against fully priced signed-Frobenius rho, its best cell is 1.566 $[0.992, 2.347]$. That interval reaches 1, but the ratio is never below 1, and the fitted gap grows as $r^{0.165\,[0.117, 0.213]}$ (benchmarks row `20261014-frobenius-orbit-ic-vs-rho`; 2,816 of 2,816 records verified).

5.2 Free relations and re-randomised targets

- *Free relations.* Self-collisions in the root table are used as homogeneous rows. In production at $n = 41, 53, 61$ this gives 0.960 [0.946, 0.976] counted and 0.941 in wall. A census closes the lever, though: at $n = 61$, $K = 600$ the self-collisions span only 49 of the 599 relations needed. Full rank from free relations alone would need $K \approx 2{,}050$ (benchmarks row `20261014-ic-next10-09-equal-advice`).
- *`stride-renew` targets.* Re-randomising the target combination at a fixed stride instead of walking it gives 0.531 [0.506, 0.557] at Koblitz $n = 13$. The result is 4.7× strong rho at $n = 19$ (benchmarks row `20261014-rootfind-stride-targets`).

5.3 Choosing the base

- *Greedy coverage.* A base chosen greedily to maximise decomposable coverage needs 4.46 expected trials instead of 5.29 on $K_0/2^{13}$, a ratio of 0.843. The gain decays with size: 1.090 at $r = 2^{16}$ and 1.0000 at $r = 2^{39}$ (`research/greedy_factor_base_yield_20261014/`). It is a real stage improvement that does not reach cryptographic sizes.
- *$n = 83$ base selection.* At Koblitz $n = 83$ the cold pipeline's base was re-selected on 16 fresh public points under a frozen protocol. Base K1107/root11 has a mean cold process of 113.755 s against 114.950 s for K1103. That is a paired reduction of 1.195 s [0.957, 1.433], and every one of the 16 per-point differences favours K1107. All 64 IC arms and 16 rho arms were admitted on a GitHub Tier 1 host, and a second local verifier reproduced the analysis exactly (`research/n83_cold_base_confirm_20261021/RESULT.md`). It is small (0.990×) but verified, and it is the current $n = 83$ cold baseline. Rho was measured online only in this study (geometric mean 2,334.964 ms), so it has no cold IC/rho ratio of its own. The last cold ratio at $n = 83$ is the earlier holdout's 55.368 (K1120, §8).
- *Kummer-tower bases over $\mathbb{F}_p$.* The μ_{2^t} bases of Petit, Kosters and Messeng [13] cost the same as a plain base of the same size to within 1% (ratios 0.999–1.012) under root finding. Their structure helps an algebraic solver, not a table (`research/pkm_tower_ecbench_20261013/`). This is a null result.

Refuted or neutral. Three bases or filters did not help:

- a cyclotomic γ-weighted base, at 1.022 [1.021, 1.023];
- a trace-zero pre-filter, at 1.00003, which is vacuous on odd order;
- symmetrised u-frame bases, which saved 0.76–0.84× at $n = 17, 23$ but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`).

Reading. Every change in this section is ENGINEERING in the constant. None changes how the relation count scales with r, and the Koblitz gap to rho still grows as roughly $r^{1/6}$ (§8). The open lever is the yield per probe at fixed N.

6 The ratchet, III: set-up and linear algebra

6.1 Set-up

Set-up is the largest term on the Koblitz pipelines (§2).

- *Stacked $n = 83$ producer* (`cryptanalysis` #938). This stacks the cryptanalysis repository's set-up optimisations into one producer. It made end-to-end cold 3.2391× [2.9809, 3.5240] faster than its own baseline. Paired within that run, the cold gap to rho fell from 305.8 to 94.4. The online ratio moved from 0.133 to 0.139, a small online regression. An integrity audit upheld the claim within a narrower scope (`crypto-autoresearcher:research/sheriff/SHF-20261016-3caa6f/findings.yaml`, C17–C19):
  - it is one run on one runner class, with no replay on another host;
  - about 1.48× of the gain is host-independent operation counts, and the rest is wall time;
  - factor-base construction is charged to neither arm;
  - the repository's "360 → 94.4" wording and a set-up field compared across rows with different meanings were judged overclaimed.

  We report the scoped figure.
- *Cache-blocked pair table.* Building the prime-field pair table in cache-sized blocks cuts set-up to 0.24 [0.22, 0.26] of the batched build and 0.13 [0.12, 0.14] of the scalar build at $N = 16{,}384$. That is 28.2 s

instead of 117 s and 224 s, and peak memory falls from 9.15 GiB to 4.65 GiB. All 456 of 456 executions are identical, and the A/A ratio is 1.02 [0.70, 1.46] (`research/ic_oracle_setup_blocked_20261020/`). Two caveats apply. The host earned L1, so the wall ratios are descriptive. And counted S is unchanged, so this is a wall-time gain that leaves the gap in the counted unit as it was (rho is 32.96, 74.31 and 269.03× cheaper at $N = 1{,}024$ to 16,384, $w = 0$).

- *u64 root batching.* Batching root extraction in 64-bit lanes gives 3.26× whole-process. The rank-stage and per-target cells were corrected on 20 October from 2.95× and 6.09× to 2.47× and 5.00×; the originals had compared the candidate's fastest run with the baseline median (benchmarks row `20261015-u64-rootbatch`).

6.2 Linear algebra

- *CADO-style filtering and Wiedemann.* On synthetic $m = 3$ matrices over a 62-bit prime, a filter followed by Wiedemann [17] beats the legacy solver by 2.48× at 2^{14} columns and 10.40× [9.99, 11.06] at 2^{18}. At 2^{18} it is 1.93× [1.74, 1.96] faster than CADO-NFS itself (`research/la_engines_20261014/`).
- *Inside ecbench.* Wired into the end-to-end pipeline, the same solver leaves counted S at 1.000 and cuts linear-algebra wall time to 0.25–0.72. Linear algebra is only 0.1–1.9% of these runs, and the best cell is still 149× rho (`research/ic_linalg_pipeline_20261019/`).
- *Deferred sparse solvers.* A negative: no deferred sparse solver lowers end-to-end S (1.206× on the $k = 4$ ladder; 5.72 → 1.030× on the scaling curve). These solvers answer only at full rank, which is a completion-rule effect, not a linear-algebra one (`research/sparse_la_timing_20261018/README.md`).
- *Sparse FGLM in the Jacobian-cover pipeline* cuts S by 1.28–1.29×. For example, IC/rho falls from 851 to 664 at $p = 53$ (`research/jv_cover_sparse_fglm_20261018/`). Both are lower bounds.

Reading. The set-up results are the largest end-to-end constant-factor gains of the period: 3.24× at $n = 83$, and 4–8× in prime set-up wall. Linear algebra is a large stage gain on a term that is small on the curves measured. It becomes the term that matters on the genus-three family (§7), where large primes and structured Gaussian elimination moved the exponent.

7 The ratchet, IV: families where the exponent moves

Two families have a fitted cost exponent below rho's. On both, the ratchet moved fastest. Neither is a deployed curve shape.

7.1 Genus three after the cubic GHS transfer

The GHS Weil descent [9] maps the discrete logarithm in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-three hyperelliptic curve over $\mathbb{F}_q$. There, double-large-prime index calculus [10] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$, against rho's $q^{3/2}$ on $E_0(\mathbb{F}_{q^3})$. The asymptotics are known. The programme contributes a priced, end-to-end, verified implementation:

- the family has $4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm;
- it runs for $q = 2^8$ to 2^{22};
- every logarithm is checked against the planted scalar;
- cost is counted in $\mathbb{F}_q$ multiplications, with inversions charged from round three on.

Five rounds, each against the previous one's frozen incumbent:

1. *Double large primes* (Ie): $q^{1.910} \to q^{1.385}$ [1.379, 1.391], and 0.036 [0.035, 0.036] of the full-base incumbent at $q = 2^{18}$ (benchmarks row `20261014-ic-improve-04-genus3-2lp`).
2. *Large primes kept as columns, structured Gaussian elimination and Wiedemann* (Ie): $q^{1.398} \to q^{1.380}$, 0.653 [0.647, 0.660] at $q = 2^{20}$.
3. *A frozen schedule $\kappa(q)$ for the factor-base fraction, plus block Wiedemann* [3] (T): 0.929 at $q = 2^{21}$; with round two, 0.605× the first double-large-prime arm. After this round a matched negation-map rho was measured at $q = 2^{20}$: IC costs $2.31 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications against rho's $2.01 \cdot 10^{10}$, i.e. 1.15× (overlapping but not identical curves).
4. *Cheaper relation primitives* (Ic): one-inversion Cantor addition, a Stickelberger pre-filter and reused exponentiations, with κ re-tuned. The result is 0.617 [0.616, 0.619] at $q = 2^{22}$, without changing a single relation (`research/ic_improve3_20261015/01-genus3-relations`).

5. *A Lucas-ladder splitting test* on the Cardano resolvent's cubic character, plus 32 walks sharing one inversion (Ie, then Ic). The per-trial log factor falls from 9.75 to 2.71 units per bit. The whole solve is 0.756 $[0.755, 0.756]$, then 0.715×, at $q = 2^{22}$; the local exponent is 1.356, and the fit over 2^{16}–2^{22} is 1.368 (`research/ic_improve4_20261015/01-genus3-relations`).

Chained, rounds three and four bring the whole solve to about 0.44× the round-two pipeline at $q = 2^{22}$. This is derived, in the inversion-charging unit.

Distance. The last matched distance is 1.15× rho at $q = 2^{20}$, after round two. Rounds three and four ran no rho arm in their unit. Whether the 0.44× carries the family across rho is therefore *not established*, and we do not claim it.

- A separate study measured IC at 0.16× a rho on $\mathrm{Jac}(C)$ at $q = 2^{14}$. That rho walks on the Jacobian, not on $E_0(\mathbb{F}_{q^3})$, so it is not the matched reference.
- The local exponents (1.36–1.41) sit above 4/3. The linear algebra grows faster than $q^{4/3}$, and re-tuning κ steepens the schedule.
- Memory now bounds the ladder: $q = 2^{22}$ peaks at 3.9 GB of a 4.5 GB cap.

7.2 $k = 4$: Gaudry's index calculus with FGHR symmetry

Over $\mathbb{F}_{p^4}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, each relation needs a four-term decomposition [8]. On a curve with rational 2-torsion T, the coordinate Y with $Y(P + T) = -Y(P)$ makes S_5 invariant under the Weyl group D_4, of order 192 [5]. The ratchet on this family:

1. *D_4 invariants* (E): the Macaulay matrix shrinks from 41,780 × 40,920 to 13,412 × 11,628. One decomposition costs 47.5–48.4× less at $p = 269$ (`research/gaudry_quartic_d4_20261013/`).
2. *D_4 Y-line on a complete ladder* (Ic): 0.0548 $[0.048, 0.061]$ of plain S_4 on $p = 101$–1033, 21 curves.
3. *Sparsest pivots, the weighted grading $\sigma_4 = \pi^2$, shape replay* (E/Ic): C_4 falls from $2.55 \cdot 10^{10}$ to $6.89 \cdot 10^7$ $\mathbb{F}_p$ multiplications (369.8× $[367.9, 371.6]$; 648 of 648 solves exact). Ecbench S falls 398–406× (`research/gaudry_quartic_d4_cost_20261014/`).
4. *First-dependency stop* (Ic): 0.979 $[0.973, 0.986]$. Double large primes cut the linear algebra to 0.827× but regress the whole solve (1.025×).
5. *A Frobenius-reduced D_4 form in characteristic two* (E/Ic): weights $(2, 2, 1, 1)$, Macaulay degree 11, a 384 × 448 matrix. It is 481× below the plain form at $m = 8$, and ecbench S is 0.0057 of the plain form at $m = 6$ (`research/gaudry_quartic_d4_char2_20261015/`).

Distance. The two characteristics are far apart.

- Over $\mathbb{F}_{p^4}$ index calculus is still $1.39 \cdot 10^6$× rho at the measured sizes.
- Over $\mathbb{F}_{(2^m)^4}$ the gap closes with size (Figure 2). IC over the negation-map rho model is 5,080 at $m = 6$ ($n \approx 2^{24}$), 648 at $m = 8$, 80.6 at $m = 10$, 49.7 at $m = 12$ and 6.06 at $m = 14$ ($n \approx 2^{56}$). Against a measured rho the last value is 4.27, from a single target. The fitted slope is -0.248 in $\log_2 n$, which is Gaudry's $n^{-1/4}$. The extrapolated crossover is near $n \approx 2^{68}$, where rho takes about 2^{34} steps, so the extrapolation can be tested directly.

After genus three, this is the nearest any index calculus in the programme has come to rho.

Joux–Vitse. An $(n - 1)$-decomposition [11] at $n = 5$ over $E(\mathbb{F}_{p^5})$ completed verified logarithms at $4.5 \cdot 10^7$–$1.8 \cdot 10^8$× rho (`research/joux_vitse_n1_20261015/`). It is a route reproduction, not an improvement.

7.3 The ledger in one table

Table 2 lists every improvement of §§4–7 that changed a default or a baseline, with its class and the distance to rho that remained after it.

8 The distance that remains

This section reports IC/rho, with greater than 1 meaning rho is cheaper. It gives the current figure for each family, the reference it uses, and how that figure moved over the period.

Table 2: The ratchet: changes that became the new incumbent, July–October 2026. Ratio is candidate/incumbent. Class against the IC incumbent: Ie exponent, Ic constant, E one stage, T trades. Boundary: St stage, Cx complete, Cd cold. "To rho" is IC/matched rho after the change, on the study's own boundary; a dash means no matched rho in that unit. Every row's IC answers were verified. "d" marks a derived reciprocal.

Change	Cl.	Bd.	Ratio [95%]	To rho
Oracle				
S_3 root finding vs matrix F4	E	St	1/1,535–1/383	—
Inherited vs matrix F4	E	St	1/222–1/199	—
`f4_fp` to msolve parity	E	St	1/54.1–1/29.8	—
Negfold root finding vs priced F4, K13	Ic	Cd	0.0545 [0.047, 0.064]	1.55× strong rho
F6-IC gate-first	E	St	0.65 [0.58, 0.75]	—
Root-rank harvest gate	Ic	Cx	1/1.20 (St 1/1.41)	—
Factor base and yield				
Free relations in production	Ic	Cx	0.960 [0.946, 0.976]	3.8–15× ($w = 0$)
`stride-renew` targets, $n = 13$	Ic	Cd	0.531 [0.506, 0.557]	4.7× at $n = 19$
N83 base K1107 vs K1103	Ic	Cd	0.990 (d; −1.195 s [0.957, 1.433])	55.4× (K1120)
Set-up and linear algebra				
N83 stacked producer	Ic	Cd	1/3.24 [1/3.52, 1/2.98]	94.4× (in-run)
Cache-blocked pair table (wall, L1)	E	St	0.24 [0.22, 0.26]	33–269×
CADO-style LA, 2^{18} columns	E	St	1/10.40 [1/11.06, 1/9.99]	149×
Genus three after cubic GHS ($\mathbb{F}_q$ multiplications)				
Double large primes	Ie	Cx	0.036; $q^{1.910} \to q^{1.385}$	2.58× model
LP columns, SGE, Wiedemann	Ie	Cx	0.653 [0.647, 0.660]	—
$\kappa(q)$ + block Wiedemann	T	Cx	0.929	1.15× measured
Relation primitives	Ic	Cx	0.617 [0.616, 0.619]	—
Lucas splitting + shared inversion	Ie/Ic	Cx	0.715; local 1.356	—
$k = 4$ with D_4 symmetry				
D_4 Y-line ladder	Ic	Cx	0.0548 [0.048, 0.061]	model handover 2^{126}–2^{137}
Pivots, weights, replay (S)	Ic	Cd	0.00249 [0.00246, 0.00251]	$1.39 \cdot 10^6$×
First-dependency stop	Ic	Cx	0.979 [0.973, 0.986]	—
Frobenius-reduced form, char. 2 (S)	Ic	Cd	0.0057	6.06× model, 4.27× measured, $n \approx 2^{56}$

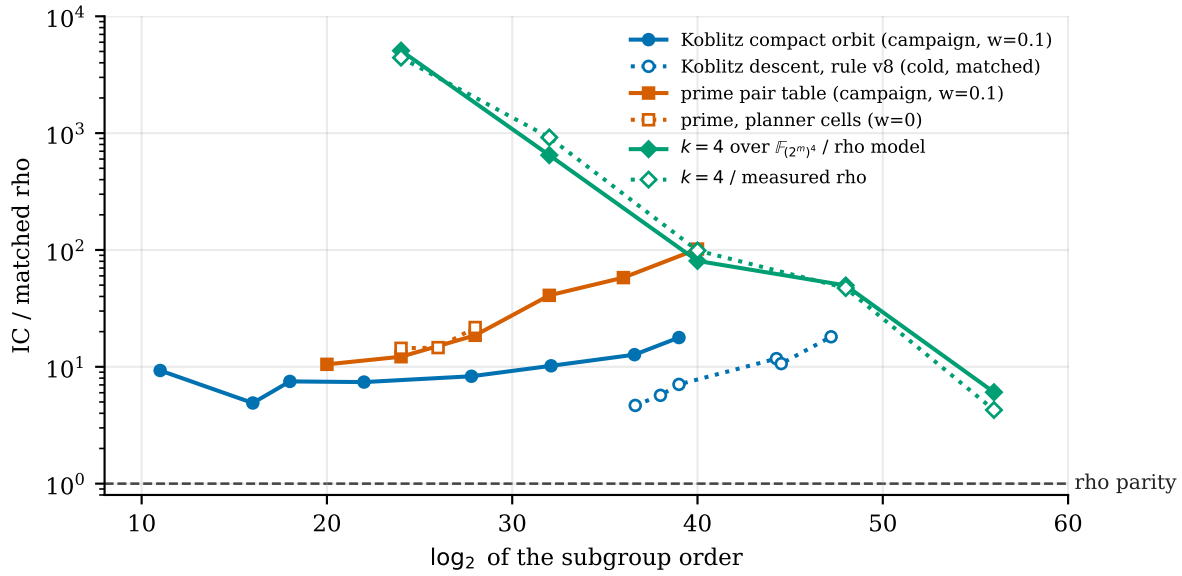


Figure 2: Distance to matched rho by family and size. Koblitz and prime curves use the frontier campaign's priced arms ($w = 0.1$, 16 targets × 2 rounds) and, open markers, the Koblitz rule at v8 (cold, 64 targets) and the prime planner cells ($w = 0$). $k = 4$ over $\mathbb{F}_{(2^m)^4}$ uses the char-2 ladder against the negation-map rho model (filled) and a measured rho (open); the $n \approx 2^{56}$ point is one target. Curves and units differ between families, so the lines are not ranked against each other. Sources: `figures/data.json`.

8.1 Koblitz curves: the rule at a matched rho

The leaderboard's Table A measures a Koblitz descent (aimed $m = 3$ collection, $m = 2$ or $m = 3$ descent) cold on one unseen public point per process: 64 points per size and six sizes from $2^{36.6}$ to $2^{47.2}$. It was re-run three times on the same instances (`research/ic_tool_program/rule/`), and Figure 3 shows all three.

- *v3 to v7: an IC improvement.* With the same rho, IC became 1.52–1.89× faster (#1978), and the cold ratio fell from 5.23–20.97 to 3.43–11.51.
- *v8: a matched reference.* The rule was run on one binary with rho's signed-Frobenius canonicalisation batched 16 lanes wide, against a same-binary control with rho as at v7. In both runs 408 of 408 rows pass the `vs_rho` check, and the matched rho takes the same walks as the control on 64 of 64 targets per size. IC's online time is unchanged between runs (0.976–1.033). Rho's online time fell 1.42–1.62×. The cold gap against the matched rho is

$$4.67\,[4.34, 5.07],\ 5.70,\ 7.07,\ 11.75,\ 10.65,\ 18.07\,[15.98, 20.58]$$

  at $\log_2 r = 36.6, 38.0, 39.0, 44.3, 44.5, 47.2$. All six values lie inside ranges declared before the rows were taken (`research/ic_tool_program/rule/v8/README.md`).

Two readings follow, and both hold.

- The 1.52–1.89× IC improvement is real and ENGINEERING.
- Against the matched rho, v8's distance sits about where v3's sat against the old rho. The fitted slope of the gap is 0.168 $[0.128, 0.208]$ per $\ln r$, against v3's 0.176. The apparent flattening between v3 and v7 was mostly an artefact of an unmatched rho. The class of v8 is ENGINEERING on the rho side and ACCOUNTING for the gap.

Pricing rho at its canonical step instead of as built gives a model gap of 6.45–23.32×. The validation sweep, recomputing Table A's rows that way, reads 9.13–41.13× (`research/validation_sweep_20261019/A-leaderboard/REPORT.md`). The leaderboard still prints the v3 ratios; the v8 figures supersede them as the distance remaining.

The priced campaign. The frontier campaign (`research/ic_frontier_campaign_20261015/`) ran every method family on the same fixed ladders, with 16 public targets per curve and every native operation priced across the band. Of its 4,848 executions, 3,178 measured runs were verified, and 19 of 19 replay audits were identical. The closest Koblitz arm, compact orbit, is 3.8–15× signed-Frobenius

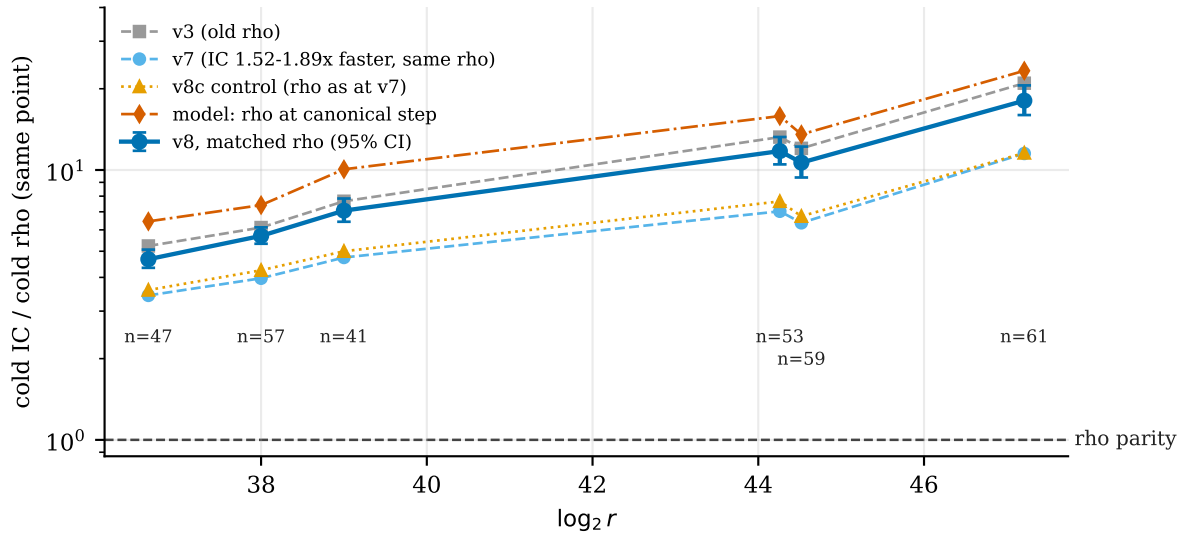


Figure 3: The Koblitz rule at three baselines, cold IC over cold rho on the same public point, 64 targets per size. v3 to v7 is an IC improvement with the same rho; v8 is the same IC against a matched (faster) rho, with a same-binary control (v8c) that reproduces v7. The dash-dot line prices rho at the canonical step. Source: `research/ic_tool_program/rule/v8/README.md`.

rho at $w = 0$ and 4.9–17.8× at $w = 0.1$. Its fitted exponent on $n = 37$–47 is 0.589 $[0.518, 0.743]$, against rho's 0.436 $[0.358, 0.508]$, separated at $w = 0$ only.

$n = 83$. The last cold measurement is the 16 October holdout with K1120, on five public points: IC at 119.826 s cold against rho's mean 2.164 s on the same points, a ratio of 55.368 (`research/n83_e2e_holdout_20261016/RESULT.md`). Its frozen selection is `BOUNDED_UNKNOWN`. K1107 since cut 1.195 s from a 115 s process (§5), which moves this ratio by about 1%. The stacked producer's in-run gap of 94.4 uses a different baseline and rho measurement (§6). Two $n = 83$ baselines in the `crypto` and `cryptanalysis` repositories (120 s and 195.7 s) are not yet reconciled.

8.2 Prime curves

- *Campaign.* The two-summand pair-table arm goes from 10.5× rho with negation at 20 bits to 101.8× at 40 bits ($w = 0.1$). Its fitted exponent is 0.620 $[0.557, 0.663]$, against rho's 0.435, which is the $r^{2/3}$ against $r^{1/2}$ of §2. The Kummer-tower arm is 28–159×.
- *Planner.* A cost-model planner chose the base and arity per curve, and was validated on six registered and three supplementary prime curves. The plan beats every alternative arm at $w = 0$, and rho is cheaper on every curve, by 12.6–25.8×. The best registered cell is 14.39 $[11.63, 18.45]$ at 24 bits, against a predicted 12.23 (1,152 runs, all verified, 72 exact replays; `research/ic_planner_validation_20261020/README.md`).
- *ECCp-131.* Exact relation paths were built for the Certicom ECCp-131 parameters, as a correctness exercise. Every arm found zero relations: a two-sum probe at $B = 512$ and a three-point probe whose hit probability is at most $2^{-106.19}$ at $B = 128$. The 20-bit ecbench control on the same code path, a different curve, is 9.41 $[7.22, 12.93]$× rho (`research/eccp131_bsgs_ic_20261020/REPORT.md`).

8.3 Nearest approaches

Table 3 collects the smallest IC/rho the programme has measured, with the qualifier each one carries. No entry is below 1 on an elliptic curve with a matched cold reference.

9 Claims that did not survive

A ratchet is worth only as much as its pawl. Several recorded results were corrected once the reference was matched, native work was priced, or the evidence was replayed. Each correction is recorded in the

Table 3: Smallest measured distances to rho, by family.

Family	IC/rho	Qualifier
Genus 3 via cubic GHS, $q = 2^{20}$	1.15	measured negation rho; overlapping, not identical curves; two later rounds (0.44×) unmatched
$k = 4$, $\mathbb{F}_{(2^m)^4}$, $m = 14$	4.27	one target; 6.06× the rho model; closing as $n^{-1/4}$
Koblitz, fully priced orbit	1.566 [0.99, 2.35]	best cell of a ladder; ratio never below 1; grows as $r^{0.165}$
Koblitz rule v8, $2^{36.6}$	4.67 [4.34, 5.07]	cold, matched, 64 targets
Prime, 20 bits	10.5	campaign, $w = 0.1$; $r^{1/6}$ widening
Prime, planner 24 bits	14.39 [11.63, 18.45]	$w = 0$
Koblitz $n = 83$	55.4	cold, K1120, `BOUNDED_UNKNOWN`

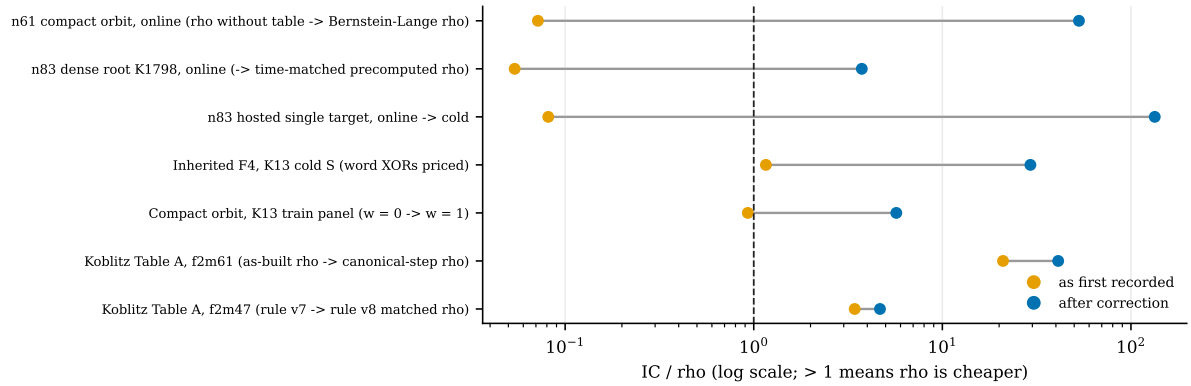


Figure 4: IC/rho as first recorded (orange) and after correction (blue), on the boundary the correction used. Values left of the dashed line had claimed IC ahead of rho. Sources: `figures/data.json`.

source with a dated note; frozen evidence is never rewritten. Figure 4 shows the before and after values. Every correction moved IC's position against rho *down*, and several moved it from apparently ahead of rho to behind it.

1. Every online win is retired. The one-target online ladder compared IC *with* a retained table against rho *without* one:

- 13.94× [9.81, 19.13] over strong rho at $n = 61$, on 64 targets;
- 18.53× at $n = 83$ with a dense root index;
- 7.29× and 12.29× on hosted single targets at $n = 71$ and $n = 83$.

Given a distinguished-point table no larger than IC's pair table, Bernstein–Lange preprocessing rho [2] is 53–317× faster online at $n = 41, 53, 61$ on the same 64 targets (benchmarks row `20261014-ic-next10-09-equal-advice`). At $n = 83$, rho with a table matched to IC's set-up time or memory wins every pair: 3.74× time-matched and 40.61× memory-matched against K3300, and 81.5–395.5× against K600 (benchmarks row `20261016-n83-precomputed-rho-control`). Cold, the hosted $n = 83$ point is 133.7× rho. Those online rows measured what a table buys, and generic rho buys more per byte. The within-IC result survives: K1798's probe reduction over K1500 at $n = 83$ (from 300,021.9 to 169,069.8 probes per relation).

2. Pricing native work moved two arms.

- *Inherited F4 at $n = 13$.* Its $S = 16.7$, close to rho's 14.4, left its Boolean word XORs unpriced. Priced, $S = 421.4$ (`research/ecbench_rootfind_faster_20261014/`).
- *F4 pricing change.* The pricing change (#1774, 0.002304 GAE per word XOR) made every earlier F4 S a lower bound. A separate algorithmic change (#1862) raised counted work (`research/f4_pricing_jump_20261018/README.md`).
- *Compact orbit at $n = 13$.* Earlier quoted at 1.08× rho as a lower bound, it reads 0.93, 1.53 and 5.7× the strong rho at $w = 0, 0.1, 1$, so the pair is unresolved inside the band (`research/ic_frontier_campaign_20261015/select/SELECTION.md`).

3. Matching the rho reference.

- *Rule v8.* Matching the rho reference raised the Koblitz gap by 1.30–1.58× at every size (§8).
- *The standing rho improved.* Ecbench's standing `rho.signed_frobenius_strong` now runs at 0.802× its former instruction count with identical GAE, and the strong Koblitz rho executes 718 instructions per step instead of 1,196. Counted-unit comparisons are unaffected. Wall-time comparisons against the older rho overstate IC by up to that factor.
- *Two rho studies on the prime side.* Choosing the better of $P \pm R$ cuts steps by 1.129×. Signed σ steps on ECC2K-130 cut GPU time to 0.961–0.970. Neither is yet in a campaign arm.

4. A replay regression, and its root cause. The root-finder ecbench session was recorded with a replay certificate of 120 of 168 identical records; the 48 mismatches were all in inherited F4's word-XOR counts. On current `main` it replayed only 72 of 168, with mismatches in all four IC arms. The validation sweep of 19–20 October isolated the cause (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`):

- *Cause.* The x86-only BMI2 word-wise rewrite in `src/cryptanalysis/inherited_f4.rs` charges a different word-XOR count from the portable path, for example 301,961 against 300,531. The method identity does not record which path ran.
- *Confirmation.* With that branch disabled, x86 reproduces all 168 of 168 records. An aarch64 build under emulation also reproduces 168 of 168. An independent rerun repeated the result.
- *What it rules out.* Compiler version, source drift since the archive, and the two later pricing changes.

Scalars, trial counts and charged GAE outside F4 are unaffected. F4 S is not comparable across instruction-set architectures unless the path is stated. Until the method identity records the path, the root-finder rows rest on the original receipt.

5. Reported cells that compared unlike statistics. The u64-rootbatch rank and per-target cells compared a fastest run with a median, and were corrected from 2.95× and 6.09× to 2.47× and 5.00× (§6). The $n = 83$ stacked producer's "360 → 94.4" cold-gap history mixed a ratio from another run; paired within the run it is 305.8 → 94.4 (§6).

6. Results resting on faulty code or wrong references.

- *Plane-quartic IC.* Its S values of 650,896 to $3.79 \cdot 10^7$ rest on a degree-3 reduction in `research/ghs_cubic_quartic_ic_20261012/quartic_ic.rs` that merges distinct classes (reported in an open pull request). Recoveries are still checked against the planted scalar, but relation counts and S are unverified.
- *Volcano census crater.* A claim that IC is 22–25% cheaper on the volcano crater was withdrawn, because the crater's rho reference should have been the cubic-automorphism rho. Against it, IC is 84.9× rho at $w = 0$ (`research/volcano_ecdlp_census_20261017/REPORT.md`, erratum A3).

Reading. None of these corrections removes an IC improvement against its own incumbent. They correct the *distance to rho*, which is the quantity most exposed to an unmatched reference or an unpriced counter. This is why the gate of §10 asks for a matched, band-priced rho on every row.

10 Validation status

As of 23 October 2026, **no index-calculus leaderboard row has passed the validation gate.** All 28 rows of the leaderboard's tables A–C render as "unvalidated (pre-gate)" (`docs/ic/leaderboard.json`, `validation_gate.status_counts`; frozen in `docs/ic/validation/PRE_GATE.json`). In `benchmarks/README.md`, 116 rows are frozen as pre-gate (`benchmarks/PRE_GATE.json`). Two rows carry a complete `VALIDATION.json`, and both are null results for IC:

- `20261017-isogeny-volcano-cost-v2`. Rho is below IC on 7,760 of 7,760 curve models, and the per-model minimum IC/rho is 17.38 at $w = 0$.
- `20261017-volcano-ecdlp-census`. IC/rho is 42.4 and 61.5 at $w = 0$ on two volcanoes, and 84.9 on the third volcano's $j = 0$ crater against the cubic-automorphism rho.

Both pass `scripts/validation_gate.py evidence`.

The repository-wide validation sweep is a screen, not the gate. It returned these verdicts on 55 leaderboard lines (`research/validation_sweep_20261019/A-leaderboard/`):

- 1 `validated`: the standing-rho Callgrind row, which is not an IC row;

- 45 `validated_with_caveats`;
- 8 `unvalidated`: the F6 ladder and the F4/F5/F6 cross-family rows, whose replay reproduced 24 of 72 records, and the retired $n = 61$ online row;
- 1 `contradicted`.

Its verdicts use a schema the gate does not read, so they change no row's standing.

What each kind of row needs. Read against §§4–8, the evidence most of the ledger lacks is clear:

- *Rows with a matched rho but no independent replay*: rule v8, the campaign, the planner and K1107. They need a second-host replay certificate and an independent validator report.
- *Rows without a matched rho in their unit*: genus three after round two, and $k = 4$ over $\mathbb{F}_{p^4}$. They need a rho run in the same binary and unit before their distance can be stated.
- *Rows whose counter is ISA-dependent*: everything with F4 word XORs. They need the method identity to record the BMI2 path before a replay can pass.
- *Stage-only rows*: oracle, set-up and linear-algebra ratios. By the gate's rule they are component sections and do not need to enter tables A–C.

Every improvement reported here is therefore a *measured, verified* improvement in the success criterion's sense: answers checked, protocol frozen before timing, receipts committed. None is yet a *validated* row in the gate's sense. We regard closing that difference as the first item of work, ahead of any new optimisation (§11).

11 The next improvements

The success criterion asks that each result name the next improvement to try. Ordered by the term of (1) that now dominates, and by how directly each would test a fitted trend:

1. *Validate what is landed.* Run the gate's three pieces of evidence on the five rows that carry most of the distance figures: rule v8, the frontier campaign, the planner cells, K1107, and the $k = 4$ characteristic-two ladder. First record the BMI2 path in F4's method identity, so that F4-bearing sessions replay across architectures.
2. *Test the $k = 4$ extrapolation.* Run the characteristic-two ladder at $m = 15$–17 with a measured negation-map rho at every rung, cold and priced across the band. The fitted $n^{-1/4}$ trend predicts parity near $n \approx 2^{68}$. A measured crossover would be the programme's first ADVANCE against rho; its absence would locate the term the model misses. Several targets per rung are needed, since the current $m = 14$ point is one target.
3. *Give genus three its distance.* Measure the matched rho on $E_0(\mathbb{F}_{q^3})$ in the unit of rounds three and four. That turns the chained $0.44\times$ into a distance, and settles whether the family crosses rho at $q \le 2^{22}$. The run past 2^{22} needs a host with more than 4.5 GB.
4. *Relation yield per probe on Koblitz curves.* None of the Koblitz changes of the period moved the $r^{1/6}$ growth of the gap. The candidates are larger m with the root-rank gate, and a base sized between the free-relation census's $K \approx 2{,}050$ and the current $K = 600$ at $n = 61$.
5. *Charge the $n = 83$ cold pipeline completely.* Run one isolated cold session with factor-base construction charged to both arms, rho cold on the same points, and the two repositories' baselines reconciled.
6. *Memory-bound exponent questions.* Two questions are blocked on memory, not on instruments: the Weil-descent cell $(m, \ell) = (4, 6)$, about 70 GB dense, and the PKM tower at $m = 3$–4, which needs 256–512 GB.

12 Conclusion

Over four months the programme made elliptic-curve index calculus much cheaper against itself.

- The decomposition oracle is two to five orders of magnitude cheaper where root finding applies.
- The Gröbner engines are one to two orders cheaper in counted work, and the $\mathbb{F}_p$ engine matches msolve.
- The $n = 83$ cold pipeline is $3.24\times$ faster after its set-up was stacked.
- Two extension-field families moved their exponents: genus three after cubic GHS from $q^{1.91}$ to a local $q^{1.36}$, and $k = 4$ with D_4 symmetry to a gap closing as $n^{-1/4}$.

By the success criterion each of these is a success.

Against rho, the honest summary is shorter. Cold, with a matched reference, rho is cheaper on every elliptic curve the programme measured:

- 4.67–18.07× on Koblitz curves of 37–47 bits;
- 10.5–102× on prime curves of 20–40 bits;
- 55× at Koblitz $n = 83$.

On the families that are not deployed shapes, the gap is 1.15× (genus three) and 4.3× ($k = 4$, one target). Every apparent win over rho in the record disappeared once the reference was given equal advice, priced alike, or matched in the same binary. Corrections of this kind lengthened the measured distance more than any single improvement shortened it.

Two things follow. First, the stage the programme improved most is no longer the stage that dominates. Relation yield per probe and table set-up now set the cost, and the next gains must come from there. Second, the nearest approaches are on families where a fitted exponent is below rho's, and both make a falsifiable prediction at sizes the harness can reach: a $k = 4$ crossover near $n \approx 2^{68}$, and a genus-three crossover at $q \leq 2^{22}$ once its rho is measured. Testing those two predictions, with every row validated, is the work that would turn the ratchet into an advance.

A Artefacts

Every number in this paper is copied from a committed artefact. The paper ran no measurement. Plotted values, each with its source, are in `papers/ecdlp-ic-improvement-ledger/figures/data.json`. The table below lists the main sources by section; the text names the rest where they are used. "Benchmarks row" means the row of that slug in `benchmarks/README.md`, whose copy of the evidence is under `benchmarks/<slug>/`.

Section	Sources
§2	`docs/ic/LEADERBOARD.md` (phase table); `research/n83_cold_base_confirm_20261021/RESULT.md`
§3	`docs/ecbench/README.md`; `docs/ic/calibration.json`; `AGENTS.md` §5a; `docs/VALIDATION_GATE.md`
§4	`RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`; `research/pdp_root_finder_20261012/`; `research/binary_mitm_oracle_l9_l10_20261015/`; `research/binary_mitm_oracle_l11_l12_20261016/`; `research/f4_f5_f6_cross_family_20261011/`; `research/f4_fp_msolve_parity_20261011/`; `research/f6_gate_first_x86_20261016/`; `research/ic_improve3_20261015/02-harvest-gate`; `research/ic_m3_algebraic_vs_mitm_20261014/`; `research/arm64_f4_defaults_20261022/`
§5	`docs/ic/COMPACT_ORBIT_PRODUCER.md`; `research/ecbench_frontier_20261013/`; `research/greedy_factor_base_yield_20261014/`; `research/n83_cold_base_confirm_20261021/`; `research/pkm_tower_ecbench_20261013/`; benchmarks rows `20261014-frobenius-orbit-ic-vs-rho`, `20261015-ic-improve2-02-freerel-production`, `20261014-rootfind-stride-targets`
§6	`crypto-autoresearcher:research/sheriff/SHF-20261016-3caa6f/findings.yaml`; `research/ic_oracle_setup_blocked_20261020/`; `research/la_engines_20261014/`; `research/ic_linalg_pipeline_20261019/`; `research/sparse_la_timing_20261018/`; `research/jv_cover_sparse_fglm_20261018/`
§7	`research/ic_improve_20261014/`; `research/ic_improve3_20261015/`; `research/ic_improve4_20261015/`; `research/gaudry_quartic_d4_20261013/`; `research/gaudry_quartic_d4_cost_20261014/`; `research/gaudry_quartic_d4_char2_20261015/`; `research/joux_vitse_n1_20261015/`
§8	`research/ic_tool_program/rule/v8/`; `research/ic_frontier_campaign_20261015/`; `research/ic_planner_validation_20261020/`; `research/n83_e2e_holdout_20261016/`; `research/eccp131_bsgs_ic_20261020/`
§9	`research/validation_sweep_20261019/`; `research/f4_pricing_jump_20261018/`; `research/volcano_ecdlp_census_20261017/`; benchmarks rows `20261014-ic-next10-09-equal-advice`, `20261016-n83-precomputed-rho-control`, `20261014-n71-n83-hosted-single-target`, `20261015-u64-rootbatch`
§10	`docs/ic/leaderboard.json`; `docs/ic/validation/PRE_GATE.json`; `benchmarks/PRE_GATE.json`; `benchmarks/20261017-isogeny-volcano-cost-v2/VALIDATION.json`; `benchmarks/20261017-volcano-ecdlp-census/VALIDATION.json`

Derived values. The paper derives a value only by taking a reciprocal or a quotient of two recorded values. Each derived value is listed here:

- every ratio written $1/s$ in Figure 1 or Table 2, which is the reciprocal of a recorded speed-up s;
- K1107 over K1103, $113.755/114.950 = 0.990$, with its interval $1 - [1.433, 0.957]/114.950$;
- the before values of Figure 4: 1/13.94, 1/18.528, 1/12.29 and 16.7/14.4, and the after value 421.4/14.4;
- the genus-three chained 0.44×, taken from the source.

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